T972

GPIO User Guide

Revision: 0.1

Release Date: 2018-12-06

Copyright

©2018 Amlogic. All rights reserved. No part of this document may be reproduced, transmitted, transcribed, or translated into any language in any form or by any means without the written permission of Amlogic.

Trademarks

Amlogic, and other Amlogic icons are trademarks of Amlogic companies. All other trademarks and registered trademarks are property of their respective holders.

Disclaimer

Amlogic may make improvements and/or changes in this document or in the product described in this document at any time.

This product is not intended for use in medical, life saving, or life sustaining applications.

Circuit diagrams and other information relating to products of Amlogic are included as a means of illustrating typical applications. Consequently, complete information sufficient for production design is not necessarily given. Amlogic makes no representations or warranties with respect to the accuracy or completeness of the contents presented in this document.

Contact Information

- Website: www.amlogic.com
- Pre-sales consultation: contact@amlogic.com
- Technical support: support@amlogic.com

Revision History

Issue 0.1 (2018-12-06)

This is the initial release.

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Contents

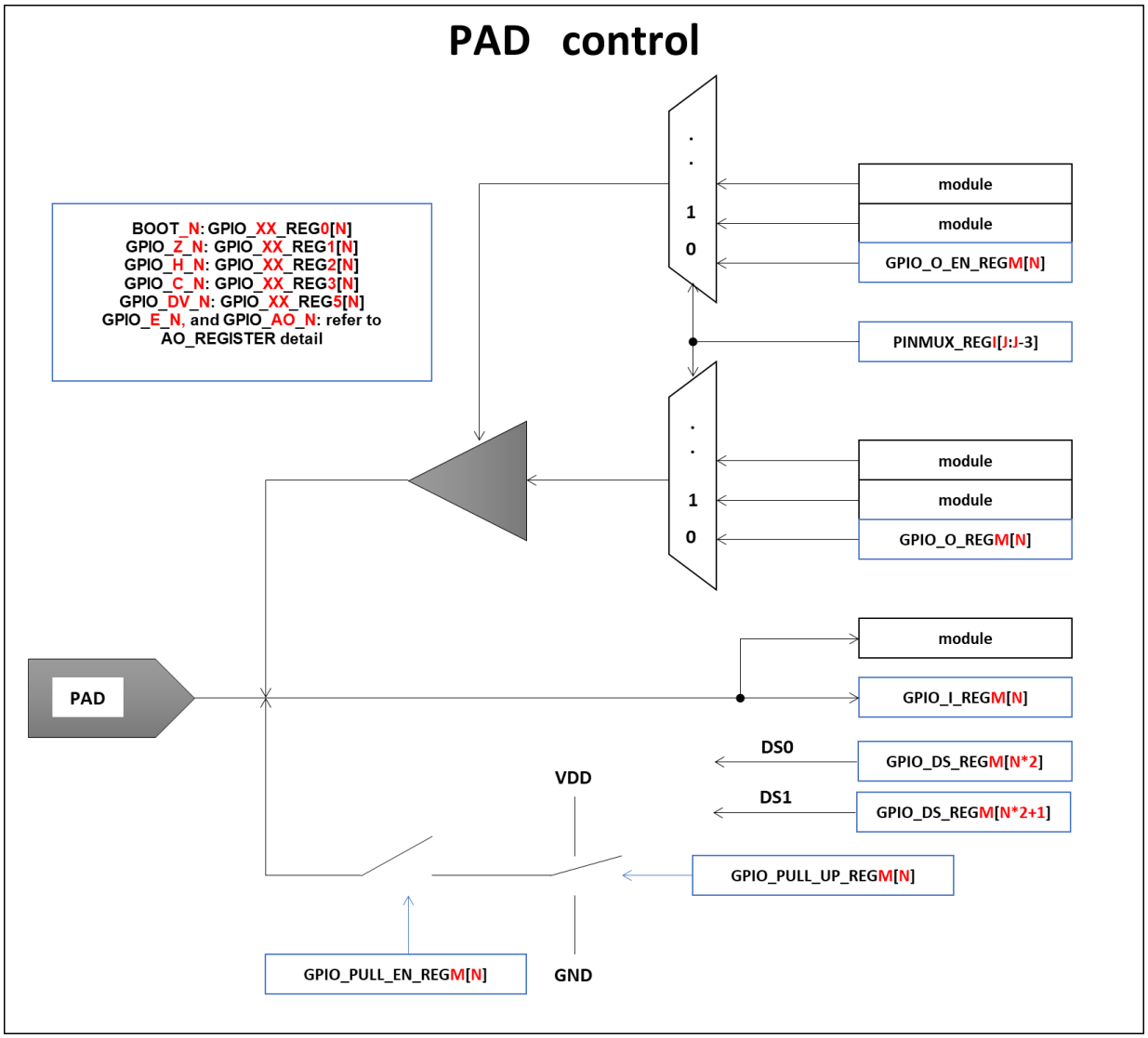
Revision History ii
1 **Overview** 1
2 **GPIO Multiplex Function** 3
3 **GPIO Interrupt** 12
4 **Register Description** 14
4.1 Pin MUX Registers 15
4.2 Pad Pull-up/down Direction 19
4.3 Pad Pull-up/down Enables 20
4.4 PREG_PAD_GPIO_I Registers 21
4.5 PREG_PAD_GPIO_O Registers 24
4.6 PREG PAD_GPIO_EN_N Registers 27
4.7 PAD_DS Registers 30
4.8 GPIOAO Registers 32

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

1 Overview

The SOC has a number of multi-function digital I/O pads that can be multiplexed to a number of internal resources (e.g. PWM generators, SDIO controllers). When a digital I/O is not being used for any specific purpose, it is converted to a general purpose GPIO pin. A GPIO pin can be statically set to high/low logical levels. The structure of a GPIO is given below.

Figure 1-1 GPIO Structure



T02FC11

Each GPIO pad has 6 related registers, among which:

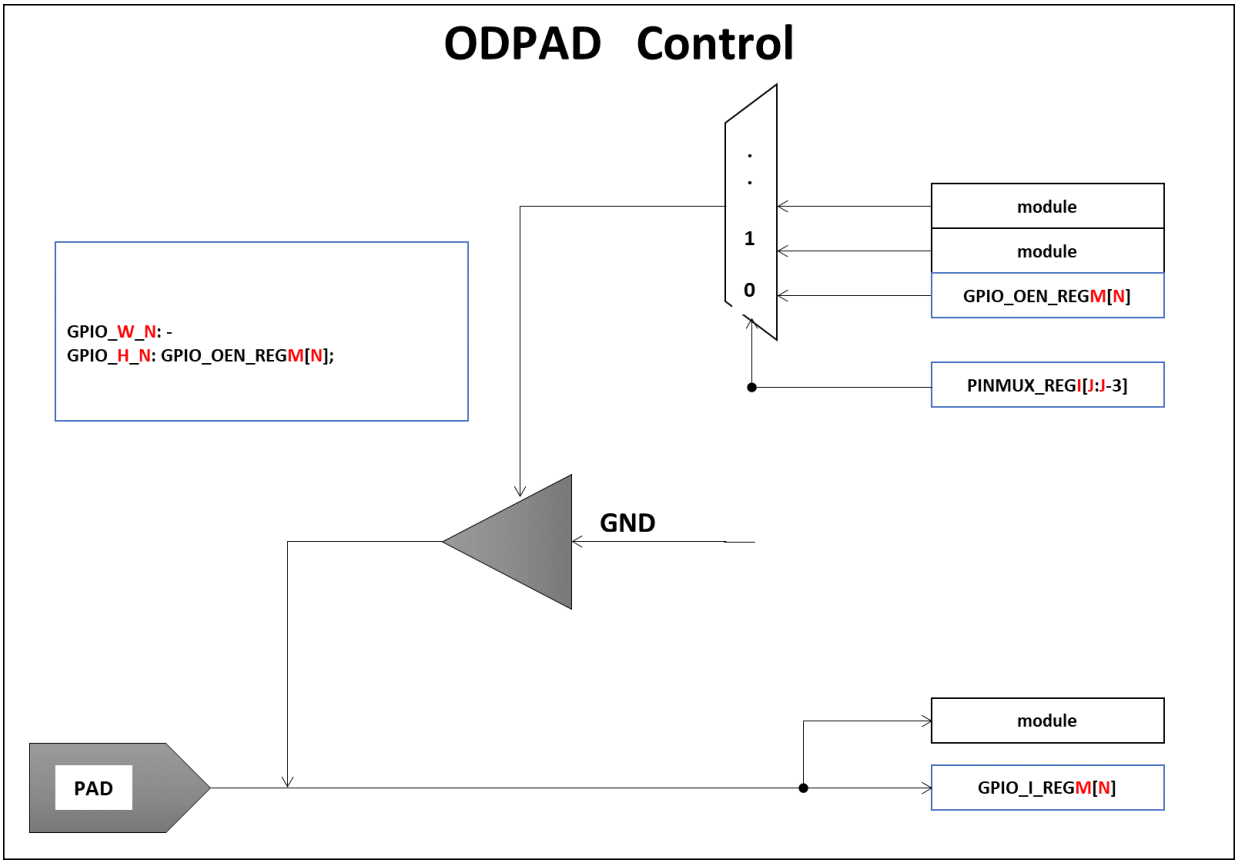
- GPIO_O_REG is used to control the output of the pad;
- GPIO_I_REG is used to store the input value of the pad;
- GPIO_O_EN_REG is used to enable GPIO output function;
- PINMUX_REG is used to define the function;
- GPIO_PULL_EN_REG is used to enable the pull-up function of GPIO pad;
- GPIO_PULL_UP_REG is used to control GPIO PAD to be whether pull-up or pull-down;

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

- For AO GPIO pads, refer to AO GPIO registers.

There are OD pins that can not be used as output, with different structure, shown as following.

Figure 1-2 OD Pad Structure



T02FC12

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

2 GPIO Multiplex Function

The GPIO multiplex functions are shown in the sections below, where the RegNN[MM] corresponds to CBUS registers defined in the following table. The base address is 0xff634400 for PERIPHS_PIN_MUX_X, PREG_PAD_GPIO_EN_N, PREG_PAD_GPIO_O, PREG_PAD_GPIO_I, PAD_PULL_UP_EN_REG, PAD_PULL_UP_REG; and 0xff800000 for AO_RTI_PIN_MUX_REG, the final address is calculated as: final address = base address + offset*4.

Table 2-1 Pin Mux Registers

Pin Mux Registers	Offset	Reset by Watchdog
AO_GPIO_I	0x0a	Yes
AO_GPIO_O	0x0d	Yes
AO_GPIO_O_EN_N	0x09	Yes
AO_RTI_PULL_UP_REG	0x0b	Yes
AO_RTI_PULL_UP_EN_REG	0x0c	Yes
AO_PAD_DS_A	0x07	Yes
AO_PAD_DS_B	0x08	Yes
AO_RTI_PINMUX_REG0	0x05	Yes
AO_RTI_PINMUX_REG1	0x06	Yes
PERIPHS_PIN_MUX_0	0xb0	Yes
PERIPHS_PIN_MUX_1	0xb1	Yes
PERIPHS_PIN_MUX_2	0xb2	Yes
PERIPHS_PIN_MUX_3	0xb3	Yes
PERIPHS_PIN_MUX_4	0xb4	Yes
PERIPHS_PIN_MUX_5	0xb5	Yes
PERIPHS_PIN_MUX_6	0xb6	Yes
PERIPHS_PIN_MUX_7	0xb7	Yes
PERIPHS_PIN_MUX_8	0xb8	Yes
PERIPHS_PIN_MUX_9	0xb9	Yes
PERIPHS_PIN_MUX_A	0xba	Yes
PERIPHS_PIN_MUX_B	0xbb	Yes
PERIPHS_PIN_MUX_C	0xbc	Yes
PERIPHS_PIN_MUX_D	0xbd	Yes
PERIPHS_PIN_MUX_E	0xbe	Yes
PERIPHS_PIN_MUX_F	0xbf	Yes
PAD_DS_REG0A	0xd0	Yes
PAD_DS_REG1A	0xd1	Yes
PAD_DS_REG2A	0xd2	Yes
PAD_DS_REG2B	0xd3	Yes

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Pin Mux Registers	Offset	Reset by Watchdog
PAD_DS_REG3A	0xd4	Yes
PAD_DS_REG4A	0xd5	Yes
PAD_DS_REG5A	0xd6	Yes
PAD_PULL_UP_REG0	0x3a	Yes
PAD_PULL_UP_REG1	0x3b	Yes
PAD_PULL_UP_REG2	0x3c	Yes
PAD_PULL_UP_REG3	0x3d	Yes
PAD_PULL_UP_REG4	0x3e	Yes
PAD_PULL_UP_REG5	0x3f	Yes
PAD_PULL_UP_EN_REG0	0x48	Yes
PAD_PULL_UP_EN_REG1	0x49	Yes
PAD_PULL_UP_EN_REG2	0x4a	Yes
PAD_PULL_UP_EN_REG3	0x4b	Yes
PAD_PULL_UP_EN_REG4	0x4c	Yes
PAD_PULL_UP_EN_REG5	0x4d	Yes
PREG_PAD_GPIO0_EN_N	0x10	Yes
PREG_PAD_GPIO0_O	0x11	Yes
PREG_PAD_GPIO0_I	0x12	Yes
PREG_PAD_GPIO1_EN_N	0x13	Yes
PREG_PAD_GPIO1_O	0x14	Yes
PREG_PAD_GPIO1_I	0x15	Yes
PREG_PAD_GPIO2_EN_N	0x16	Yes
PREG_PAD_GPIO2_O	0x17	Yes
PREG_PAD_GPIO2_I	0x18	Yes
PREG_PAD_GPIO3_EN_N	0x19	Yes
PREG_PAD_GPIO3_O	0x1a	Yes
PREG_PAD_GPIO3_I	0x1b	Yes
PREG_PAD_GPIO4_EN_N	0x1c	Yes
PREG_PAD_GPIO4_O	0x1d	Yes
PREG_PAD_GPIO4_I	0x1e	Yes
PREG_PAD_GPIO5_EN_N	0x20	Yes
PREG_PAD_GPIO5_O	0x21	Yes
PREG_PAD_GPIO5_I	0x22	Yes
AO_GPIO_TEST_N	0xd7	No

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 2-2 GPIOAO_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5
GPIO-AO_0	AO_RTI_ PINMUX_REG0	[3:0]	UART_AO_A_ TX				
GPIO-AO_1		[7:4]	UART_AO_A_ RX				
GPIO-AO_2		[11:-8]	I2C_AO_M0_ SCL	UART_AO_ B_TX	I2C_AO_S0_ SCL		
GPIO-AO_3		[15:-12]	I2C_AO_M0_ SDA	UART_AO_ B_RX	I2C_AO_S0_ SDA		
GPIO-AO_4		[19:-16]	IR_REMOTE_ OUT	CLK_32K_IN	PWMAO_C	PWMAO_C_ HIZ	
GPIO-AO_5		[23:-20]	IR_REMOTE_ INPUT		PWMAO_D		
GPIO-AO_6		[27:-24]	JTAG_A_CLK	SPDIF_OUT	PWMAO_C		
GPIO-AO_7		[31:-28]	JTAG_A_TMS	SPDIF_IN	PWMAO_C_ HIZ		
GPIO-AO_8	AO_RTI_ PINMUX_REG1	[3:0]	JTAG_A_TDI		UART_AO_ B_TX		
GPIO-AO_9		[7:4]	JTAG_A_TDO	IR_ REMOTE_ OUT	UART_AO_ B_RX		
GPIO-AO_ 10		[11:-8]	AO_CEC_A	AO_CEC_B	PWMAO_D		
GPIO-AO_11		[15:-12]		PWMAO_A_ HIZ	PWMAO_A	GEN_CLK_ EE	GEN_CLK_ AO

Table 2-3 GPIOE_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4
GPIO-E_0	AO_RTI_PINMUX_ REG1	[19:1-6]	UART_AO_A_ CTS	UART_AO_B_ CTS	PWMAO_B	I2C_AO_M0_ SCL
GPIO-E_1		[23:2-0]	UART_AO_A_ RTS	UART_AO_B_ RTS	PWMAO_D	I2C_AO_M0_ SDA
GPIO-E_2		[27:2-4]	CLK12_24	CLK25_EE	PWM_A	

Table 2-4 BOOT_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3
BOOT_0	PERIPHS_PIN_MUX_ 0	[3:0]	EMMC_D0		

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Pin Name	Control Register	Contr-ol Bit	Fun1	Func2	Func3
BOOT_1		[7:4]	EMMC_D1		
BOOT_2		[11:8]	EMMC_D2		
BOOT_3		[15:12]	EMMC_D3		SPIF_HOLD
BOOT_4		[19:16]	EMMC_D4		SPIF_MO
BOOT_5		[23:20]	EMMC_D5		SPIF_MI
BOOT_6		[27:24]	EMMC_D6		SPIF_CLK
BOOT_7		[31:28]	EMMC_D7		SPIF_WP
BOOT_8	PERIPHS_PIN_MUX_1	[3:0]	EMMC_CLK	NAND_WEN_CLK	
BOOT_9		[7:4]		NAND_ALE	
BOOT_10		[11:8]	EMMC_CMD	NAND_REN_WR	
BOOT_11		[15:12]	EMMC_DS	NAND_CLE	
BOOT_12		[19:16]		NAND_CE0	
BOOT_13		[23:20]			SPIF_CS

Table 2-5 GPIODV_x Multi-Function Pin

Pin Na-me	Control Register	C-o-nt-rol Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GP-IO-D-V_0	PERIPHS_PIN_MUX_C	[3:-0]	I2C_EE_M0_SDA	TCON_SLV_SDA	TSin_D7_A			LVDS_PROD_TEST_O_22	DEBUG_BUS_OUT0
GP-IO-D-V_1		[7:-4]	I2C_EE_M0_SCL	TCON_SLV_SCK	TSin_D6_A			LVDS_PROD_TEST_O_23	DEBUG_BUS_OUT1
GP-IO-D-V_2		[1-1:-8]	DTV_IF_AGC	ATV_IF_AGC	TSin_D5_A	TCON_AGING	PWM_C	LVDS_PROD_TEST_O_24	DEBUG_BUS_OUT2
GP-IO-D-V_3		[1-5:-12]	DTV_IF_AGC	DTV_RF_AGC	TSin_D4_A		PWM_D		DEBUG_BUS_OUT3
GP-IO-D-V_4		[1-9:-16]	SPDIF_OUT	TCON_FLICKER	TSin_D3_A	IS-O7816_CLK			DEBUG_BUS_OUT4
GP-IO-D-V_5		[2-3:-20]	SPDIF_IN		TSin_D2_A	IS-O7816_DATA			DEBUG_BUS_OUT5

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIO_DV_6		[27:24]	TDMC_D3	TDMC_DIN3	TSin_D1_A	PWM_E			DEBUG_BUS_OUT6
GPIO_DV_7		[31:28]	TDMC_SCLK	TDMC_SLV_SCLK	TSin_D0_A	SPI_MISO_B	UART_EE_C_TX		DEBUG_BUS_OUT7
GPIO_DV_8	PERIPHS_PIN_MUX_D	[3:0]	TDMC_FS	TDMC_SLV_FS	TSin_CLK_A	SPI_MOSI_B	UART_EE_C_RX		DEBUG_BUS_OUT8
GPIO_DV_9		[7:4]	TDMC_D0	TDMC_DIN0	TSin_SOP_A	SPI_CLK_B	UART_EE_C_CTS	I2C_EE_M3_SCL	DEBUG_BUS_OUT9
GPIO_DV_10		[11:8]	TDMC_D1	TDMC_DIN1	TSin_VALID_A	SPI_SS0_B	UART_EE_C_RTS	I2C_EE_M3_SDA	DEBUG_BUS_OUT10
GPIO_DV_11		[15:12]	TDMC_D2	TDMC_DIN2	SPDIF_OUT	SPI_SS1_B	PWM_F		DEBUG_BUS_OUT11

Table 2-6 GPIOW_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Fun2
GPIOW_0	PERIPHS_PIN_MUX_A	[3:0]	HDMIRX_HPD_A	
GPIOW_1		[7:4]	HDMIRX_5VDET_A	
GPIOW_2		[11:8]	HDMIRX_SDA_A	UART_AO_A_TX
GPIOW_3		[15:12]	HDMIRX_SCL_A	UART_AO_A_RX
GPIOW_4		[19:16]	HDMIRX_HPD_C	
GPIOW_5		[23:20]	HDMIRX_5VDET_C	
GPIOW_6		[27:24]	HDMIRX_SDA_C	UART_AO_A_TX
GPIOW_7		[31:28]	HDMIRX_SCL_C	UART_AO_A_RX
GPIOW_8	PERIPHS_PIN_MUX_B	[3:0]	HDMIRX_HPD_B	
GPIOW_9		[7:4]	HDMIRX_5VDET_B	
GPIOW_10		[11:8]	HDMIRX_SDA_B	UART_AO_A_TX
GPIOW_11		[15:12]	HDMIRX_SCL_B	UART_AO_A_RX

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 2-7 GPIOH_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOH_0	PERIPH-S_PIN_MUX_7	[3:0]	TCON_0	SPI_SS1_A	HSYNC	ETH_ACT_LED	SPDIF_OUT	I2C_EE_M3_SCL	ETH_DEBUG_IN0
GPIOH_1		[7:4]	TCON_1	SPI_SS2_A	VSYNC	ETH_LINK_LED		I2C_EE_M3_SDA	ETH_DEBUG_IN1
GPIOH_2		[11:8]	TCON_2	ETH_MDIO	BT656_A_CLK	TCON_SFC			ETH_DEBUG_IN2
GPIOH_3		[15:12]	TCON_3	ETH_MDC	BT656_A_VS				ETH_DEBUG_IN3
GPIOH_4		[19:16]	TCON_4	ETH_RGMII_RX_CLK	BT656_A_HS	TSin_D0_B	MCLK_0		ETH_DEBUG_IN4
GPIOH_5		[23:20]	TCON_5	ETH_RX_DV	BT656_A_DIN0	TSin_CLK_B	TDMA_SCLK		ETH_DEBUG_IN5
GPIOH_6		[27:24]	TCON_6	ETH_RXD0	BT656_A_DIN1	TSin_SOP_B	TDMA_FS		ETH_DEBUG_IN6
GPIOH_7		[31:28]	TCON_7	ETH_RXD1	BT656_A_DIN2	TSin_VALID_B	TDMA_D0		ETH_DEBUG_IN7
GPIOH_8	PERIPH-S_PIN_MUX_8	[3:0]	TCON_8	ETH_RXD2_RGMII	BT656_A_DIN3	TSin_D0_C	TDMA_D1		ETH_DEBUG_IN8
GPIOH_9		[7:4]	TCON_9	ETH_RXD3_RGMII	BT656_A_DIN4	TSin_CLK_C	TDMA_D2		ETH_DEBUG_IN9
GPIOH_10		[11:8]	TCON_10	ETH_RGMII_TX_CLK	BT656_A_DIN5	TSin_SOP_C	TDMA_D3		ETH_DEBUG_IN10
GPIOH_11		[15:12]	TCON_11	ETH_TXEN	BT656_A_DIN6	TSin_VALID_C	TDMA_DIN0		ETH_DEBUG_IN11
GPIOH_12		[19:16]	TCON_12	ETH_TXD0	BT656_A_DIN7		TDMA_DIN1		ETH_DEBUG_IN12
GPIOH_13		[23:20]	TCON_13	ETH_TXD1	CLK12_24	PWM_C	TDMA_DIN2		ETH_DEBUG_IN13
GPIOH_14		[27:24]	TCON_14	ETH_TXD2_RGMII	3D_SYNC_OUT		TDMA_DIN3		
GPIOH_15		[31:28]	TCON_15	ETH_TXD3_RGMII	VX1_HTPDN				

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOH_16	PERIPH-S_PIN_MUX_9	[3:0]	TCON_LOCK	TCON_SFC	VX1_LOCKN				
GPIOH_17		[7:4]	TCON_SPI_MO	SPI_MISO_A		UART_EE_B_TX	GEN_CLK_EE		DEBUG_BUS_OUT12
GPIOH_18		[11:8]	TCON_SPI_MI	SPI_MOSI_A		UART_EE_B_RX			DEBUG_BUS_OUT13
GPIOH_19		[15:12]	TCON_SPI_CLK	SPI_CLK_A	I2C_EE_M2_SCL	UART_EE_B_CTS	I2C_EE_M3_SCL		DEBUG_BUS_OUT14
GPIOH_20		[19:16]	TCON_SPI_SS	SPI_SS0_A	I2C_EE_M2_SDA	UART_EE_B_RTS	I2C_EE_M3_SDA		DEBUG_BUS_OUT15
GPIOH_21		[23:20]	I2C_EE_M1_SCL				GEN_CLK_EE		DEBUG_BUS_OUT16
GPIOH_22		[27:24]	I2C_EE_M1_SDA						DEBUG_BUS_OUT17

Table 2-8 GPIOC_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOC_0	PERIPH-S_PIN_MUX_4	[3:0]	SDCAR-D_D0	JTAG_B_TDO	TDMB_D0	TSout_CLK	SPI_MISO_B		
GPIOC_1		[7:4]	SDCAR-D_D1	JTAG_B_TDI	TDMB_D1	TSout_SOP	SPI_MOSI_B		
GPIOC_2		[11:8]	SDCAR-D_D2	UART_AO_A_RX	TDMB_D2	TSout_VALID	SPI_CLK_B	LVDS_PROD_TEST_O_0	TST_CLK_OUT0
GPIOC_3		[15:12]	SDCAR-D_D3	UART_AO_A_TX	TDMB_D3	TSout_D0	SPI_SS0_B	LVDS_PROD_TEST_O_1	TST_CLK_OUT1
GPIOC_4		[19:16]	SDCAR-D_CLK	JTAG_B_CLK	TDMB_SCLK	TSout_D1	SPI_SS1_B	LVDS_PROD_TEST_O_2	TST_CLK_OUT2
GPIOC_5		[23:20]	SDCAR-D_CMD	JTAG_B_TMS	TDMB_FS	TSout_D2	SPI_SS2_B	LVDS_PROD_TEST_O_3	TST_CLK_OUT3
GPIOC_6		[27:24]	UART_EE_A_TX	PDM_DCLK	TDMB_DIN0	TSout_D3	DTV_IF_AGC	ATV_IF_AGC	TST_CLK_OUT4
GPIOC_7		[31:28]	UART_EE_A_RX	PDM_DIN0	TDMB_DIN1	TSout_D4	DTV_IF_AGC	DTV_RF_AGC	TST_CLK_OUT5

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func5	Func6	Func7
GPIOC_8	PERIPH-S_PIN_MUX_5	[3:0]	UART_EE_A_CTS	PDM_DIN1	TDMB_DIN2	TSout_D5	I2C_EE_M0_SCL	LVDS_PROD_TEST_O_4	TST_CLK_OUT6
GPIOC_9		[7:4]	UART_EE_A_RTS	PDM_DIN2	TDMB_DIN3	TSout_D6	I2C_EE_M0_SDA	LVDS_PROD_TEST_O_5	TST_CLK_OUT7
GPIOC_10		[11:8]	PWM_B	PDM_DIN3	MCLK_1	TSout_D7		LVDS_PROD_TEST_O_6	TST_CLK_OUT8
GPIOC_11		[15:12]	SPI_MISO_A	TDMB_D1	TDMB_DIN1			LVDS_PROD_TEST_O_7	TST_CLK_OUT9
GPIOC_12		[19:16]	SPI_MOSI_A	TDMB_D0	TDMB_DIN0	WIFI_BEACO-N		LVDS_PROD_TEST_O_8	TST_CLK_OUT10
GPIOC_13		[23:20]	SPI_CLK_A	TDMB_SCLK	TDMB_SLV_SCLK	I2C_EE_M3_SCL		LVDS_PROD_TEST_O_9	TST_CLK_OUT11
GPIOC_14		[27:24]	SPI_SS0_A	TDMB_FS	TDMB_SLV_FS	I2C_EE_M3_SDA		LVDS_PROD_TEST_O_10	TST_CLK_OUT12

Table 2-9 GPIOZ_x Multi-Function Pin

Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func6	Func7
GPIOZ_0	PERIPH-S_PIN_MUX_2	[3:0]	MCLK_0				LVDS_PROD_TEST_O_11	TST_CLK_OUT13
GPIOZ_1		[7:4]	TDMA_SCLK	TDMA_SLV_SCLK	I2C_EE_M1_SDA		LVDS_PROD_TEST_O_12	TST_CLK_OUT14
GPIOZ_2		[11:8]	TDMA_FS	TDMA_SLV_FS	I2C_EE_M1_SCL		LVDS_PROD_TEST_O_13	TST_CLK_OUT15
GPIOZ_3		[15:12]	TDMA_D0			PWM_A	LVDS_PROD_TEST_O_14	TST_CLK_OUT16
GPIOZ_4		[19:16]	TDMA_D1		WIFI_BEACON	PWM_B	LVDS_PROD_TEST_O_15	TST_CLK_OUT17

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

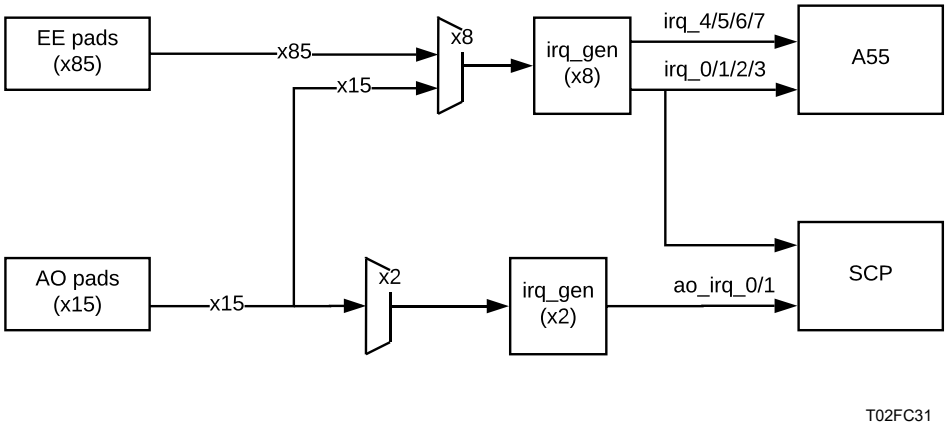
Pin Name	Control Register	Control Bit	Fun1	Func2	Func3	Func4	Func6	Func7
GPIOZ_5		[23:20]	TDMA_D2		PWM_VS	PWM_C	LVDS_PROD_TEST_O_16	TST_CLK_OUT18
GPIOZ_6		[27:24]	TDMA_D3		PWM_VS	PWM_D	LVDS_PROD_TEST_O_17	TST_CLK_OUT19
GPIOZ_7		[31:28]	SPDIF_OUT	TDMA_DIN0	PDM_DCLK		LVDS_PROD_TEST_O_18	
GPIOZ_8	PERIPH-S_PIN_MUX_3	[3:0]	MCLK_1	TDMA_DIN1	PDM_DIN0		LVDS_PROD_TEST_O_19	
GPIOZ_9		[7:4]	I2C_EE_M2_SCL	TDMA_DIN2	PDM_DIN1	PWM_E	LVDS_PROD_TEST_O_20	
GPIOZ_10		[11:8]	I2C_EE_M2_SDA	TDMA_DIN3	PDM_DIN2	PWM_F	LVDS_PROD_TEST_O_21	

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

3 GPIO Interrupt

There are 10 interrupts that are routed to all CPUs (A55/SCP). The tables below outline the interrupts associated with each interrupt group.

Figure 3-1 GPIO Interrupt



There are 8 independent filtered GPIO interrupt modules that can be programmed to use any of the GPIOs in the chip as an interrupt source (listed in the table below). For example, to select gpioZ_15 as the source for GPIO IRQ #0, then CBUS 0x2609 bits[7:0] = 27 (according to the table below).

Input Mux Location CBUS registers 0x2621 and 0x2622	**Description**
[255:102]	0
[101:99]	gpioE[2:0]
[98:87]	gpioDV[11:0]
[86:75]	gpioW[11:0]
[74:60]	gpioC[14:0]
[59:46]	boot[13:0]
[45:23]	gpioH[22:0]
[22:12]	gpioZ[10:0]
[11:0]	gpioAO[11:0]

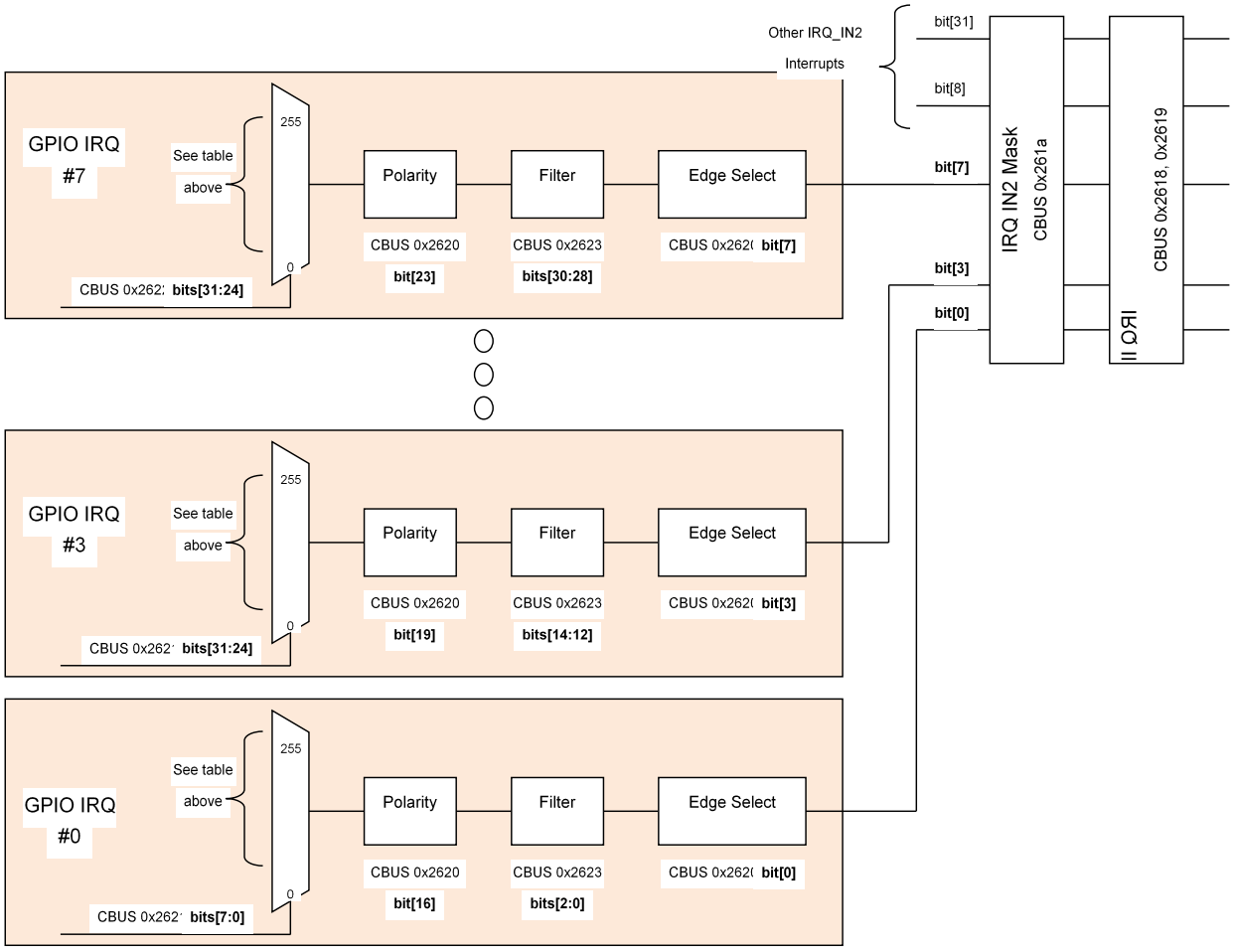
The diagram below illustrates the path a GPIO takes to become an interrupt. The eight GPIO interrupts respond to the MASK, STATUS and STATUS/CLEAR registers just like any other interrupt in the chip. The difference for the GPIO interrupts is that they can be filtered and conditioned.

Note

The input for the GPIO interrupt module (the input into the 256:1 mux) comes directly from the I/O pad of the chip. Therefore if a pad (say gpioA_11) is configured as a UART TX pin, then in theory, the UART TX pin can be a GPIO interrupt since the TX pin will drive gpioA_11 which in turn can drive the GPIO interrupt module.

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Figure 3-2 GPIO Interrupt Path



T02FC32

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

4 Register Description

The following table shows the mapping information of GPIO I/O Registers and Nets.

Register	Offset	Net Name
PREG_PAD_GPIO0_EN_N	0x10	BOOT
PREG_PAD_GPIO0_O	0x11	
PREG_PAD_GPIO0_I	0x12	
PAD_PULL_UP_REG0	0x3a	
PAD_PULL_UP_EN_REG0	0x48	
PAD_DS_REG0A	0xd0	
PERIPHS_PIN_MUX_0	0xb0	
PERIPHS_PIN_MUX_1	0xb1	
PREG_PAD_GPIO3_EN_N	0x19	GPIOC
PREG_PAD_GPIO3_O	0x1a	
PREG_PAD_GPIO3_I	0x1b	
PAD_PULL_UP_REG3	0x3d	
PAD_PULL_UP_EN_REG3	0x4b	
PERIPHS_PIN_MUX_4	0xb4	
PERIPHS_PIN_MUX_5	0xb5	
PREG_PAD_GPIO2_EN_N	0x16	GPIOH
PREG_PAD_GPIO2_O	0x17	
PREG_PAD_GPIO2_I	0x18	
PAD_PULL_UP_REG2	0x3c	
PAD_PULL_UP_EN_REG2	0x4a	
PAD_DS_REG2A	0xd2	
PAD_DS_REG2B	0xd3	
PAD_DS_REG3A	0xd4	
PERIPHS_PIN_MUX_7	0xb7	
PERIPHS_PIN_MUX_8	0xb8	
PERIPHS_PIN_MUX_9	0xb9	
PREG_PAD_GPIO4_EN_N	0x1c	GPIOW
PREG_PAD_GPIO4_O	0x1d	
PREG_PAD_GPIO4_I	0x1e	
PERIPHS_PIN_MUX_A	0xba	
PERIPHS_PIN_MUX_B	0xbb	
PREG_PAD_GPIO5_EN_N	0x20	GPIODV

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Register	Offset	Net Name
PREG_PAD_GPIO5_O	0x21	
PREG_PAD_GPIO5_I	0x22	
PAD_PULL_UP_REG5	0x3f	
PAD_PULL_UP_EN_REG5	0x4d	
PAD_DS_REG5A	0xd6	
PERIPHS_PIN_MUX_C	0xbc	
PERIPHS_PIN_MUX_D	0xbd	
PAD_PULL_UP_REG1	0x3b	GPIOZ
PAD_PULL_UP_EN_REG1	0x49	
PAD_DS_REG1A	0xd1	
PERIPHS_PIN_MUX_2	0xb2	
PERIPHS_PIN_MUX_3	0xb3	
PREG_PAD_GPIO1_EN_N	0x13	
PREG_PAD_GPIO1_O	0x14	
PREG_PAD_GPIO1_I	0x15	

4.1 Pin MUX Registers

Table 4-1 PERIPHS_PIN_MUX_0 0xb0

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for BOOT_0
[7:4]	R/W	0	pin mux select bits for BOOT_1
[11:8]	R/W	0	pin mux select bits for BOOT_2
[15:12]	R/W	0	pin mux select bits for BOOT_3
[19:16]	R/W	0	pin mux select bits for BOOT_4
[23:20]	R/W	0	pin mux select bits for BOOT_5
[27:24]	R/W	0	pin mux select bits for BOOT_6
[31:28]	R/W	0	pin mux select bits for BOOT_7

Table 4-2 PERIPHS_PIN_MUX_1 0xb1

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for BOOT_8
[7:4]	R/W	0	pin mux select bits for BOOT_9
[11:8]	R/W	0	pin mux select bits for BOOT_10
[15:12]	R/W	0	pin mux select bits for BOOT_11

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
[19:16]	R/W	0	pin mux select bits for BOOT_12
[23:20]	R/W	0	pin mux select bits for BOOT_13

Table 4-3 PERIPHS_PIN_MUX_2 0xb2

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOZ_0
[7:4]	R/W	0	pin mux select bits for GPIOZ_1
[11:8]	R/W	0	pin mux select bits for GPIOZ_2
[15:12]	R/W	0	pin mux select bits for GPIOZ_3
[19:16]	R/W	0	pin mux select bits for GPIOZ_4
[23:20]	R/W	0	pin mux select bits for GPIOZ_5
[27:24]	R/W	0	pin mux select bits for GPIOZ_6
[31:28]	R/W	0	pin mux select bits for GPIOZ_7

Table 4-4 PERIPHS_PIN_MUX_3 0xb3

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOZ_8
[7:4]	R/W	0	pin mux select bits for GPIOZ_9
[11:8]	R/W	0	pin mux select bits for GPIOZ_10

Table 4-5 PERIPHS_PIN_MUX_4 0xb4

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOC_0
[7:4]	R/W	0	pin mux select bits for GPIOC_1
[11:8]	R/W	0	pin mux select bits for GPIOC_2
[15:12]	R/W	0	pin mux select bits for GPIOC_3
[19:16]	R/W	0	pin mux select bits for GPIOC_4
[23:20]	R/W	0	pin mux select bits for GPIOC_5
[27:24]	R/W	0	pin mux select bits for GPIOC_6
[31:28]	R/W	0	pin mux select bits for GPIOC_7

Table 4-6 PERIPHS_PIN_MUX_5 0xb5

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOC_8
[7:4]	R/W	0	pin mux select bits for GPIOC_9

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
[11:8]	R/W	0	pin mux select bits for GPIOC_10
[15:12]	R/W	0	pin mux select bits for GPIOC_11
[19:16]	R/W	0	pin mux select bits for GPIOC_12
[23:20]	R/W	0	pin mux select bits for GPIOC_13
[27:24]	R/W	0	pin mux select bits for GPIOC_14

PERIPHS_PIN_MUX_6 0xb6

Reserved.

Table 4-7 PERIPHS_PIN_MUX_7 0xb7

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_0
[7:4]	R/W	0	pin mux select bits for GPIOH_1
[11:8]	R/W	0	pin mux select bits for GPIOH_2
[15:12]	R/W	0	pin mux select bits for GPIOH_3
[19:16]	R/W	0	pin mux select bits for GPIOH_4
[23:20]	R/W	0	pin mux select bits for GPIOH_5
[27:24]	R/W	0	pin mux select bits for GPIOH_6
[31:28]	R/W	0	pin mux select bits for GPIOH_7

Table 4-8 PERIPHS_PIN_MUX_8 0xb8

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_8
[7:4]	R/W	0	pin mux select bits for GPIOH_9
[11:8]	R/W	0	pin mux select bits for GPIOH_10
[15:12]	R/W	0	pin mux select bits for GPIOH_11
[19:16]	R/W	0	pin mux select bits for GPIOH_12
[23:20]	R/W	0	pin mux select bits for GPIOH_13
[27:24]	R/W	0	pin mux select bits for GPIOH_14
[31:28]	R/W	0	pin mux select bits for GPIOH_15

Table 4-9 PERIPHS_PIN_MUX_9 0xb9

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOH_16
[7:4]	R/W	0	pin mux select bits for GPIOH_17
[11:8]	R/W	0	pin mux select bits for GPIOH_18

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
[15:12]	R/W	0	pin mux select bits for GPIOH_19
[19:16]	R/W	0	pin mux select bits for GPIOH_20
[23:20]	R/W	0	pin mux select bits for GPIOH_21
[27:24]	R/W	0	pin mux select bits for GPIOH_22

Table 4-10 PERIPHS_PIN_MUX_A 0xba

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOW_0
[7:4]	R/W	0	pin mux select bits for GPIOW_1
[11:8]	R/W	0	pin mux select bits for GPIOW_2
[15:12]	R/W	0	pin mux select bits for GPIOW_3
[19:16]	R/W	0	pin mux select bits for GPIOW_4
[23:20]	R/W	0	pin mux select bits for GPIOW_5
[27:24]	R/W	0	pin mux select bits for GPIOW_6
[31:28]	R/W	0	pin mux select bits for GPIOW_7

Table 4-11 PERIPHS_PIN_MUX_B 0xbb

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIOW_8
[7:4]	R/W	0	pin mux select bits for GPIOW_9
[11:8]	R/W	0	pin mux select bits for GPIOW_10
[15:12]	R/W	0	pin mux select bits for GPIOW_11

Table 4-12 PERIPHS_PIN_MUX_C 0xbc

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIODV_0
[7:4]	R/W	0	pin mux select bits for GPIODV_1
[11:8]	R/W	0	pin mux select bits for GPIODV_2
[15:12]	R/W	0	pin mux select bits for GPIODV_3
[19:16]	R/W	0	pin mux select bits for GPIODV_4
[23:20]	R/W	0	pin mux select bits for GPIODV_5
[27:24]	R/W	0	pin mux select bits for GPIODV_6
[31:28]	R/W	0	pin mux select bits for GPIODV_7

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-13 PERIPHS_PIN_MUX_D 0xbd

Bit(s)	R/W	Default	Description
[3:0]	R/W	0	pin mux select bits for GPIODV_8
[7:4]	R/W	0	pin mux select bits for GPIODV_9
[11:8]	R/W	0	pin mux select bits for GPIODV_10
[15:12]	R/W	0	pin mux select bits for GPIODV_11

PERIPHS_PIN_MUX_E 0xbe

Reserved.

PERIPHS_PIN_MUX_F 0xbf

Reserved.

4.2 Pad Pull-up/down Direction

The I/O pads contain both a pull-up and a pull-down. If a bit is set to 1 in the registers below, then the pull-up is enabled. If a bit is set to 0, then the pull-down is enabled.

Note

There are separate pull-up “enables” that must also be set to 1 for the pull-up/down direction to function. If an “enable” is set to 0, then the pull-up/down feature is disabled, and the bits below are ignored (on a per pad basis).

Table 4-14 PAD_PULL_UP_REG0 0x3a

Bit(s)	R/W	Default	Description
31~14	R/W	0x35ff	Unused
13~0	R/W		Boot[13:0] 1 = pull up. 0 = pull down

Table 4-15 PAD_PULL_UP_REG1 0x3b

Bit(s)	R/W	Default	Description
31~11	R/W	0x60e	Unused
10~0	R/W		gpioZ[10:0] 1 = pull up. 0 = pull down

Table 4-16 PAD_PULL_UP_REG2 0x3c

Bit(s)	R/W	Default	Description
31~23	R/W	0x0000-00	Reserved
22~17	R/W		gpioH[22:17] 1 = pull up. 0 = pull down
16	R/W		Reserved
15~0	R/W		gpioH[15:0] 1 = pull up. 0 = pull down

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-17 PAD_PULL_UP_REG3 0x3d

Bit(s)	R/W	Default	Description
31~15	R/W	0x57ff	Reserved
14~0	R/W		gpioC[14:0] 1 = pull up. 0 = pull down

PAD_PULL_UP_REG4 0x3e

Reserved

Table 4-18 PAD_PULL_UP_REG5 0x3f

Bit(s)	R/W	Default	Description
31~12	R/W	0xfc7	Unused
11~0	R/W		gpioDV[11:0] 1 = pull up. 0 = pull down

4.3 Pad Pull-up/down Enables

Each I/O pad has a selectable pull-up or pull-down resistor. For the pull-up direction (up or down) to be operational, the appropriate bit below must be set in order to enable the pull-up/down function.

Table 4-19 PAD_PULL_UP_EN_REG0 0x48

Bit(s)	R/W	Default	Description
31~14	R/W	0x3fff	Unused
13~0	R/W		boot[13:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Table 4-20 PAD_PULL_UP_EN_REG1 0x49

Bit(s)	R/W	Default	Description
31~11	R/W	0x7ff	Unused
10~0	R/W		gpioZ[10:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Table 4-21 PAD_PULL_UP_EN_REG2 0x4a

Bit(s)	R/W	Default	Description
31~23	R/W	0x00ffff	Reserved
22~17	R/W		gpioH[22:17] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down
16	R/W		Reserved
15~0	R/W		gpioH[15:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-22 PAD_PULL_UP_EN_REG3 0x4b

Bit(s)	R/W	Default	Description
31~15	R/W	0x7fff	reserved
14~0	R/W		gpioC[14:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

PAD_PULL_UP_EN_REG4 0x4c

Reserved

Table 4-23 PAD_PULL_UP_EN_REG5 0x4d

Bit(s)	R/W	Default	Description
31~12	R/W	0xfff	reserved
11~0	R/W		gpioDV[11:0] pullup-enable: 1 = pullup or pull-down enabled. 0 = no-pull-up or pull-down

4.4 PREG_PAD_GPIO_I Registers

Table 4-24 PREG_PAD_GPIO0_I 0x12

Bit(s)	R/W	Default	Description
0	R/W	0	BOOT_0 select
1	R/W	0	BOOT_1 select
2	R/W	0	BOOT_2 select
3	R/W	0	BOOT_3 select
4	R/W	0	BOOT_4 select
5	R/W	0	BOOT_5 select
6	R/W	0	BOOT_6 select
7	R/W	0	BOOT_7 select
8	R/W	0	BOOT_8 select
9	R/W	0	BOOT_9 select
10	R/W	0	BOOT_10 select
11	R/W	0	BOOT_11 select
12	R/W	0	BOOT_12 select
13	R/W	0	BOOT_13 select

Table 4-25 PREG_PAD_GPIO1_I 0x15

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOZ_0 select
1	R/W	0	GPIOZ_1 select
2	R/W	0	GPIOZ_2 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
3	R/W	0	GPIOZ_3 select
4	R/W	0	GPIOZ_4 select
5	R/W	0	GPIOZ_5 select
6	R/W	0	GPIOZ_6 select
7	R/W	0	GPIOZ_7 select
8	R/W	0	GPIOZ_8 select
9	R/W	0	GPIOZ_9 select
10	R/W	0	GPIOZ_10 select

Table 4-26 PREG_PAD_GPIO2_I 0x18

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOH_0 select
1	R/W	0	GPIOH_1 select
2	R/W	0	GPIOH_2 select
3	R/W	0	GPIOH_3 select
4	R/W	0	GPIOH_4 select
5	R/W	0	GPIOH_5 select
6	R/W	0	GPIOH_6 select
7	R/W	0	GPIOH_7 select
8	R/W	0	GPIOH_8 select
9	R/W	0	GPIOH_9 select
10	R/W	0	GPIOH_10 select
11	R/W	0	GPIOH_11 select
12	R/W	0	GPIOH_12 select
13	R/W	0	GPIOH_13 select
14	R/W	0	GPIOH_14 select
15	R/W	0	GPIOH_15 select
16	R/W	0	GPIOH_16 select
17	R/W	0	GPIOH_17 select
18	R/W	0	GPIOH_18 select
19	R/W	0	GPIOH_19 select
20	R/W	0	GPIOH_20 select
21	R/W	0	GPIOH_21 select
22	R/W	0	GPIOH_22 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-27 PREG_PAD_GPIO3_I 0x1b

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOC_0 select
1	R/W	0	GPIOC_1 select
2	R/W	0	GPIOC_2 select
3	R/W	0	GPIOC_3 select
4	R/W	0	GPIOC_4 select
5	R/W	0	GPIOC_5 select
6	R/W	0	GPIOC_6 select
7	R/W	0	GPIOC_7 select
8	R/W	0	GPIOC_8 select
9	R/W	0	GPIOC_9 select
10	R/W	0	GPIOC_10 select
11	R/W	0	GPIOC_11 select
12	R/W	0	GPIOC_12 select
13	R/W	0	GPIOC_13 select
14	R/W	0	GPIOC_14 select

Table 4-28 PREG_PAD_GPIO4_I 0x1e

Bit(s)	R/W	Default	Description
0	R/W	0	GPIOW_0 select
1	R/W	0	GPIOW_1 select
2	R/W	0	GPIOW_2 select
3	R/W	0	GPIOW_3 select
4	R/W	0	GPIOW_4 select
5	R/W	0	GPIOW_5 select
6	R/W	0	GPIOW_6 select
7	R/W	0	GPIOW_7 select
8	R/W	0	GPIOW_8 select
9	R/W	0	GPIOW_9 select
10	R/W	0	GPIOW_10 select
11	R/W	0	GPIOW_11 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-29 PREG_PAD_GPIO5_I 0x22

Bit(s)	R/W	Default	Description
0	R/W	0	GPIODV_0 select
1	R/W	0	GPIODV_1 select
2	R/W	0	GPIODV_2 select
3	R/W	0	GPIODV_3 select
4	R/W	0	GPIODV_4 select
5	R/W	0	GPIODV_5 select
6	R/W	0	GPIODV_6 select
7	R/W	0	GPIODV_7 select
8	R/W	0	GPIODV_8 select
9	R/W	0	GPIODV_9 select
10	R/W	0	GPIODV_10 select
11	R/W	0	GPIODV_11 select

4.5 PREG_PAD_GPIO_O Registers

Table 4-30 PREG_PAD_GPIO0_O 0x11

Bit(s)	R/W	Default	Description
0	R/W	1	BOOT_0 select
1	R/W	1	BOOT_1 select
2	R/W	1	BOOT_2 select
3	R/W	1	BOOT_3 select
4	R/W	1	BOOT_4 select
5	R/W	1	BOOT_5 select
6	R/W	1	BOOT_6 select
7	R/W	1	BOOT_7 select
8	R/W	1	BOOT_8 select
9	R/W	1	BOOT_9 select
10	R/W	1	BOOT_10 select
11	R/W	1	BOOT_11 select
12	R/W	1	BOOT_12 select
13	R/W	1	BOOT_13 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-31 PREG_PAD_GPIO1_O 0x14

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOZ_0 select
1	R/W	1	GPIOZ_1 select
2	R/W	1	GPIOZ_2 select
3	R/W	1	GPIOZ_3 select
4	R/W	1	GPIOZ_4 select
5	R/W	1	GPIOZ_5 select
6	R/W	1	GPIOZ_6 select
7	R/W	1	GPIOZ_7 select
8	R/W	1	GPIOZ_8 select
9	R/W	1	GPIOZ_9 select
10	R/W	1	GPIOZ_10 select

Table 4-32 PREG_PAD_GPIO2_O 0x17

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOH_0 select
1	R/W	1	GPIOH_1 select
2	R/W	1	GPIOH_2 select
3	R/W	1	GPIOH_3 select
4	R/W	1	GPIOH_4 select
5	R/W	1	GPIOH_5 select
6	R/W	1	GPIOH_6 select
7	R/W	1	GPIOH_7 select
8	R/W	1	GPIOH_8 select
9	R/W	1	GPIOH_9 select
10	R/W	1	GPIOH_10 select
11	R/W	1	GPIOH_11 select
12	R/W	1	GPIOH_12 select
13	R/W	1	GPIOH_13 select
14	R/W	1	GPIOH_14 select
15	R/W	1	GPIOH_15 select
16	R/W	1	NA
17	R/W	1	GPIOH_17 select
18	R/W	1	GPIOH_18 select
19	R/W	1	GPIOH_19 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
20	R/W	1	GPIOH_20 select
21	R/W	1	GPIOH_21 select
22	R/W	1	GPIOH_22 select

Table 4-33 PREG_PAD_GPIO3_O 0x1a

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOC_0 select
1	R/W	1	GPIOC_1 select
2	R/W	1	GPIOC_2 select
3	R/W	1	GPIOC_3 select
4	R/W	1	GPIOC_4 select
5	R/W	1	GPIOC_5 select
6	R/W	1	GPIOC_6 select
7	R/W	1	GPIOC_7 select
8	R/W	1	GPIOC_8 select
9	R/W	1	GPIOC_9 select
10	R/W	1	GPIOC_10 select
11	R/W	1	GPIOC_11 select
12	R/W	1	GPIOC_12 select
13	R/W	1	GPIOC_13 select
14	R/W	1	GPIOC_14 select

PREG_PAD_GPIO4_O 0x1d

Reserved.

Table 4-34 PREG_PAD_GPIO5_O 0x21

Bit(s)	R/W	Default	Description
0	R/W	1	GPIODV_0 select
1	R/W	1	GPIODV_1 select
2	R/W	1	GPIODV_2 select
3	R/W	1	GPIODV_3 select
4	R/W	1	GPIODV_4 select
5	R/W	1	GPIODV_5 select
6	R/W	1	GPIODV_6 select
7	R/W	1	GPIODV_7 select
8	R/W	1	GPIODV_8 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
9	R/W	1	GPIODV_9 select
10	R/W	1	GPIODV_10 select
11	R/W	1	GPIODV_11 select

4.6 PREG PAD_GPIO_EN_N Registers

Table 4-35 PREG_PAD_GPIO0_EN_N 0x10

Bit(s)	R/W	Default	Description
0	R/W	1	BOOT_0 select
1	R/W	1	BOOT_1 select
2	R/W	1	BOOT_2 select
3	R/W	1	BOOT_3 select
4	R/W	1	BOOT_4 select
5	R/W	1	BOOT_5 select
6	R/W	1	BOOT_6 select
7	R/W	1	BOOT_7 select
8	R/W	1	BOOT_8 select
9	R/W	1	BOOT_9 select
10	R/W	1	BOOT_10 select
11	R/W	1	BOOT_11 select
12	R/W	1	BOOT_12 select
13	R/W	1	BOOT_13 select

Table 4-36 PREG_PAD_GPIO1_EN_N 0x13

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOZ_0 select
1	R/W	1	GPIOZ_1 select
2	R/W	1	GPIOZ_2 select
3	R/W	1	GPIOZ_3 select
4	R/W	1	GPIOZ_4 select
5	R/W	1	GPIOZ_5 select
6	R/W	1	GPIOZ_6 select
7	R/W	1	GPIOZ_7 select
8	R/W	1	GPIOZ_8 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
9	R/W	1	GPIOZ_9 select
10	R/W	1	GPIOZ_10 select

Table 4-37 PREG_PAD_GPIO2_EN_N 0x16

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOH_0 select
1	R/W	1	GPIOH_1 select
2	R/W	1	GPIOH_2 select
3	R/W	1	GPIOH_3 select
4	R/W	1	GPIOH_4 select
5	R/W	1	GPIOH_5 select
6	R/W	1	GPIOH_6 select
7	R/W	1	GPIOH_7 select
8	R/W	1	GPIOH_8 select
9	R/W	1	GPIOH_9 select
10	R/W	1	GPIOH_10 select
11	R/W	1	GPIOH_11 select
12	R/W	1	GPIOH_12 select
13	R/W	1	GPIOH_13 select
14	R/W	1	GPIOH_14 select
15	R/W	1	GPIOH_15 select
16	R/W	1	GPIOH_16 select
17	R/W	1	GPIOH_17 select
18	R/W	1	GPIOH_18 select
19	R/W	1	GPIOH_19 select
20	R/W	1	GPIOH_20 select
21	R/W	1	GPIOH_21 select
22	R/W	1	GPIOH_22 select

Table 4-38 PREG_PAD_GPIO3_EN_N 0x19

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOC_0 select
1	R/W	1	GPIOC_1 select
2	R/W	1	GPIOC_2 select
3	R/W	1	GPIOC_3 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
4	R/W	1	GPIOC_4 select
5	R/W	1	GPIOC_5 select
6	R/W	1	GPIOC_6 select
7	R/W	1	GPIOC_7 select
8	R/W	1	GPIOC_8 select
9	R/W	1	GPIOC_9 select
10	R/W	1	GPIOC_10 select
11	R/W	1	GPIOC_11 select
12	R/W	1	GPIOC_12 select
13	R/W	1	GPIOC_13 select
14	R/W	1	GPIOC_14 select

Table 4-39 PREG_PAD_GPIO4_EN_N 0x1c

Bit(s)	R/W	Default	Description
0	R/W	1	GPIOW_0 select
1	R/W	1	GPIOW_1 select
2	R/W	1	GPIOW_2 select
3	R/W	1	GPIOW_3 select
4	R/W	1	GPIOW_4 select
5	R/W	1	GPIOW_5 select
6	R/W	1	GPIOW_6 select
7	R/W	1	GPIOW_7 select
8	R/W	1	GPIOW_8 select
9	R/W	1	GPIOW_9 select
10	R/W	1	GPIOW_10 select
11	R/W	1	GPIOW_11 select

Table 4-40 PREG_PAD_GPIO5_EN_N 0x20

Bit(s)	R/W	Default	Description
0	R/W	1	GPIODV_0 select
1	R/W	1	GPIODV_1 select
2	R/W	1	GPIODV_2 select
3	R/W	1	GPIODV_3 select
4	R/W	1	GPIODV_4 select
5	R/W	1	GPIODV_5 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
6	R/W	1	GPIODV_6 select
7	R/W	1	GPIODV_7 select
8	R/W	1	GPIODV_8 select
9	R/W	1	GPIODV_9 select
10	R/W	1	GPIODV_10 select
11	R/W	1	GPIODV_11 select

4.7 PAD_DS Registers

Table 4-41 PAD_DS_REG0A 0xd0

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	BOOT_0 select
[3:2]	R/W		BOOT_1 select
[5:4]	R/W		BOOT_2 select
[7:6]	R/W		BOOT_3 select
[9:8]	R/W		BOOT_4 select
[11:10]	R/W		BOOT_5 select
[13:12]	R/W		BOOT_6 select
[15:14]	R/W		BOOT_7 select
[17:16]	R/W		BOOT_8 select
[19:18]	R/W		BOOT_9 select
[21:20]	R/W		BOOT_10 select
[23:22]	R/W		BOOT_11 select
[25:24]	R/W		BOOT_12 select
[27:26]	R/W		BOOT_13 select

Table 4-42 PAD_DS_REG1A 0xd1

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOZ_0 select
[3:2]	R/W		GPIOZ_1 select
[5:4]	R/W		GPIOZ_2 select
[7:6]	R/W		GPIOZ_3 select
[9:8]	R/W		GPIOZ_4 select
[11:10]	R/W		GPIOZ_5 select
[13:12]	R/W		GPIOZ_6 select

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Bit(s)	R/W	Default	Description
[15:14]	R/W		GPIOZ_7 select
[17:16]	R/W		GPIOZ_8 select
[19:18]	R/W		GPIOZ_9 select
[21:20]	R/W		GPIOZ_10 select

Table 4-43 PAD_DS_REG2A 0xd2

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOH_0 select
[3:2]	R/W		GPIOH_1 select
[5:4]	R/W		GPIOH_2 select
[7:6]	R/W		GPIOH_3 select
[9:8]	R/W		GPIOH_4 select
[11:10]	R/W		GPIOH_5 select
[13:12]	R/W		GPIOH_6 select
[15:14]	R/W		GPIOH_7 select
[17:16]	R/W		GPIOH_8 select
[19:18]	R/W		GPIOH_9 select
[21:20]	R/W		GPIOH_10 select
[23:22]	R/W		GPIOH_11 select
[25:24]	R/W		GPIOH_12 select
[27:26]	R/W		GPIOH_13 select
[29:28]	R/W		GPIOH_14 select
[31:30]	R/W		GPIOH_15 select

Table 4-44 PAD_DS_REG2B 0xd3

Bit(s)	R/W	Default	Description
[3:2]	R/W	2'b10	GPIOH_17
[5:4]	R/W		GPIOH_18
[7:6]	R/W		GPIOH_19
[9:8]	R/W		GPIOH_20
[11:10]	R/W		GPIOH_21
[13:12]	R/W		GPIOH_22

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-45 PAD_DS_REG3A 0xd4

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIOC_0
[3:2]	R/W		GPIOC_1
[5:4]	R/W		GPIOC_2
[7:6]	R/W		GPIOC_3
[9:8]	R/W		GPIOC_4
[11:10]	R/W		GPIOC_5
[13:12]	R/W		GPIOC_6
[15:14]	R/W		GPIOC_7
[17:16]	R/W		GPIOC_8
[19:18]	R/W		GPIOC_9
[21:20]	R/W		GPIOC_10
[23:22]	R/W		GPIOC_11
[25:24]	R/W		GPIOC_12
[27:26]	R/W		GPIOC_13
[29:28]	R/W		GPIOC_14

PAD_DS_REG4A 0xd5

Reserved.

Table 4-46 PAD_DS_REG5A 0xd6

Bit(s)	R/W	Default	Description
[1:0]	R/W	2'b10	GPIODV_0
[3:2]	R/W		GPIODV_1
[5:4]	R/W		GPIODV_2
[7:6]	R/W		GPIODV_3
[9:8]	R/W		GPIODV_4
[11:10]	R/W		GPIODV_5
[13:12]	R/W		GPIODV_6
[15:14]	R/W		GPIODV_7
[17:16]	R/W		GPIODV_8
[19:18]	R/W		GPIODV_9
[21:20]	R/W		GPIODV_10
[23:22]	R/W		GPIODV_11

4.8 GPIOAO Registers

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Always On GPIO Input levels

Table 4-47 AO_GPIO_I 0x0a

Bit(s)	R/W	Default	Description
31:19	R	0	reserved
18:16	R	0	INPUT_LEVELS: These bits correspond to the INPUT levels on the gpioE[2:0] pins
15:12	R	0	reserved
11-0	R	0	INPUT_LEVELS: These bits correspond to the INPUT levels on the gpioAO[11:0] pins

Table 4-48 AO_GPIO_O 0x0d

Bit(s)	R/W	Default	Description
31	R/W	1	reserved
18-16	R/W	0xFFF	OUTPUT_LEVEL: These bits correspond to the output levels on the gpioE[2:0] pins when in GPIO mode.
11-0	R/W	0xFFF	OUTPUT_LEVEL: These bits correspond to the output levels on the gpioAO [11:0] pins when in GPIO mode.

Table 4-49 AO_RTI_PIN_MUX_REG0 0x05

Bit(s)	R/W	Default	Description
31-0	R/W	0	Pin mux settings

Table 4-50 AO_RTI_PIN_MUX_REG1 0x06

Bit(s)	R/W	Default	Description
31-0	R/W	0	Pin mux settings

Table 4-51 AO_GPIO_O_EN_N 0x09

Bit(s)	R/W	Default	Description
30-19	R	0	Reserved
18-16	R/W	0x7	OUTPUT_ENABLE: These bits correspond to the output levels on the gpioE[2:0] pins when in GPIO mode. A ‘0’ sets the gpioE pin to be an output.
15-12	R	0	Reserved
11-0	R/W	0xfff	OUTPUT_ENABLE: These bits correspond to the output levels on the gpioAO [11:0] pins when in GPIO mode. A ‘0’ sets the gpioAO pin to be an output.

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-52 AO_RTI_PULL_UP_REG 0x0b

Bit(s)	R/W	Default	Description
31	R/W	1	Reserved.
18-16	R/W	0x3dff	gpioE[2:0] pull-up/down. 1 = pull-up 0 = pull-down
11-0	R/W	0x9df	gpioAO[11:0] pull up/down. 1 = pull-up 0 = pull-down

Table 4-53 AO_RTI_PULL_UP_EN_REG 0x0c

Bit(s)	R/W	Default	Description
31	R/W	1	Reserved.
18-16	R/W	0x3dff	gpioE[2:0] pull-up/down enable. 1 = pull-up/down enable, 0 = pull-up/down disable
11-0	R/W	0x9df	gpioAO[11:0] pull-up/down enable. 1 = pull-up/down enable, 0 = pull-up/down disable

Pad ds0/ds1

Table 4-54 AO_PAD_DS_A 0x07

Bit(s)	R/W	Default	Description
23:22	R/W	2'b10	gpioAO_11{ds1 ds0}
21:20	R/W		gpioAO_10{ds1 ds0}
19:18	R/W		gpioAO_9{ds1 ds0}
17:16	R/W		gpioAO_8{ds1 ds0}
15:14	R/W		gpioAO_7{ds1 ds0}
13:12	R/W		gpioAO_6{ds1 ds0}
11:10	R/W		gpioAO_5{ds1 ds0}
9:8	R/W		gpioAO_4{ds1 ds0}
7:6	R/W		gpioAO_3{ds1 ds0}
5:4	R/W		gpioAO_2{ds1 ds0}
3:2	R/W		gpioAO_1{ds1 ds0}
1:0	R/W		gpioAO_0 {ds1 ds0}

Pad ds0/ds1

Table 4-55 AO_PAD_DS_B 0x08

Bit(s)	R/W	Default	Description
31:30	R/W	2'b10	reserved
5:4	R/W		gpioE_2{ds1 ds0}
3:2	R/W		gpioE_1{ds1 ds0}
1:0	R/W		gpioE_0{ds1 ds0}

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.

Table 4-56 AO_PINMUX_LOCK 0x17

Bit(s)	R/W	Default	Description
31	R/W	0	1: generate error when write to each locked register;
7	R/W	0	lock AO gpio_o
6	R/W	0	lock AO gpio pull en
5	R/W	0	lock AO gpio pull up/down
4	R/W	0	lock AO gpio_oen_n
1	R/W	0	lock ao pin_mux_reg1
0	R/W	0	lock ao pin_mux_reg0

Always On GPIO controls.

Caution

This register is NOT reset during a watchdog event.

Table 4-57 AO_GPIO_TEST_N 0xd7

Bit(s)	R/W	Default	Description
31	R/W	1	TEST_N_OUTPUT_LEVEL: This bit controls the output level of the test_n pin when TEST_N_GPIO_EN is set to 1.
30	R	0	TEST_N input level
29-12	R/W	0	reserved
11-8	R/W	0	testn_sel
7-6	R/W	0	reserved
5	R/W	1	test_n_pull_up_en
4	R/W	1	test_n_pull_up
3-2	R/W	2’b10	test_n_ds
1-0	R/W	2’b10	reset_n_ds

Amlogic Proprietary and Confidential
Copyright © Amlogic. All rights reserved.